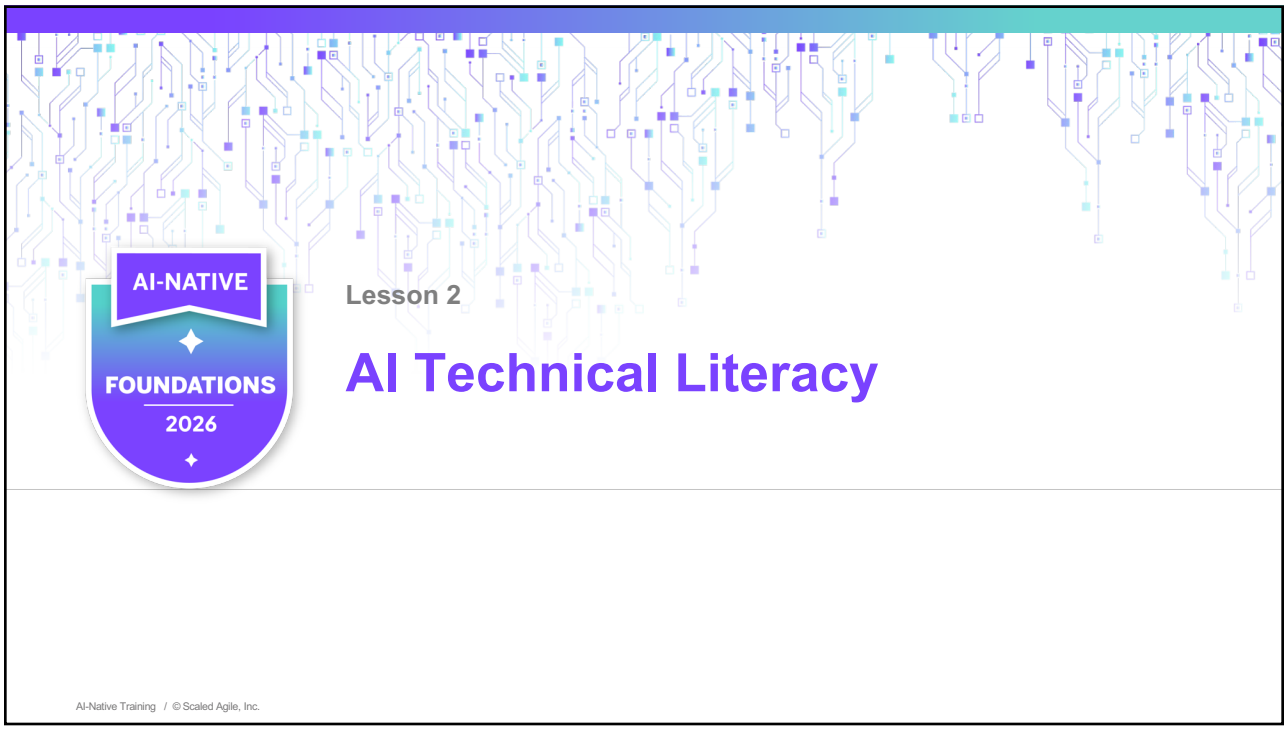


1

Learning Objectives

By the end of this lesson, you will:

- Navigate common AI terminology
- Understand how Large Language Models work
- Craft effective prompts
- Apply the principles of Responsible AI
- Describe Retrieval-Augmented Generation (RAG)
- Understand how AI Agents work and their capabilities
- Classify AI use cases as Stable, Evolving, or Frontier

2

2.1 AI Basics

3

Class Schedule

Day 1

Lesson 1: EDGE & AI-Native

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Lesson 6: Roadmap to AI-Native

Lesson 7: Tell Your AI-Native Story

4

Explain to Someone Who Doesn't Know About AI

5 min

Pair share:

Scenario: How would you explain AI to someone who doesn't know what it is?

- **Instructions:**
  - Pair up with someone near you
  - Each person gets one to two minutes to explain
  - Make it as simple and clear as possible
  - Your listener notes anything confusing or overly technical
  - Give your partner feedback

5

Machine vs. Human Intelligence



Narrow and task-specific

Excels at finding patterns in large datasets

Executes those patterns at incredible scale

Lacks consciousness, intent, or ethical judgment





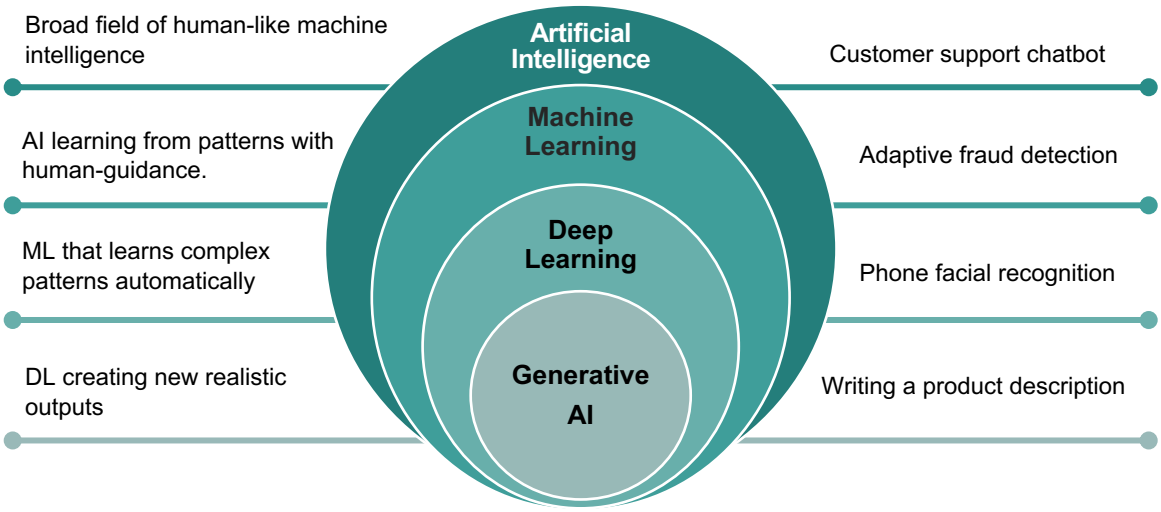
Flexible and emotionally aware

Capable of abstraction and moral reasoning

Can understand context and nuance

6

Understanding AI, ML, DL, and GenAI



7

Understand The Increasing Levels Of Intelligence

Think of it like this:
AI is like going from grade school to a PhD—starting with memorization (grade school), building understanding through experience and analysis (high school), connecting complex ideas (university), and finally creating something entirely new (Ph.D.).

8

One Problem, Three Approaches

7 min

Individual Activity

- **Step 1:** Pick a real-world problem (e.g., traffic prediction, fraud detection, customer service).
- **Step 2:** Ask the LLM: *"Explain how this problem could be solved with Machine Learning, Deep Learning, and Generative AI."*
- **Step 3:** Discuss the similarities and differences with how different types of AI handle the problem.

9

Debrief

3 min

Class discussion

- What problem did you choose, and what was the most interesting difference between the three AI solutions?
- Based on the problems shared, when is a predictive approach (ML/DL) better than a generative one?

10

Insights and Action

1 min

Reflect

Before this session, how would you have explained AI to someone who knew nothing about it? Now write down one key distinction between human and machine intelligence that you'd emphasize differently. What surprised you most about how AI, ML, deep learning, and generative AI relate to each other?

Apply

Ask AI: "Here are five technologies I use regularly: Netflix recommendations, iPhone Face ID, voice-to-text, spam email filtering, and Grammarly writing suggestions. Help me categorize each as AI, Machine Learning, Deep Learning, or Generative AI. For each one, explain the input, what the system learns, and the output it produces."

Explore

MIT's Generative AI Explained—foundational AI concepts clarified

Stanford 2025 AI Index Report—comprehensive AI landscape guide

MIT Sloan AI Basics Guide—practical AI applications today

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2.2 Data

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Data as a Design Decision

3 min

Class discussion

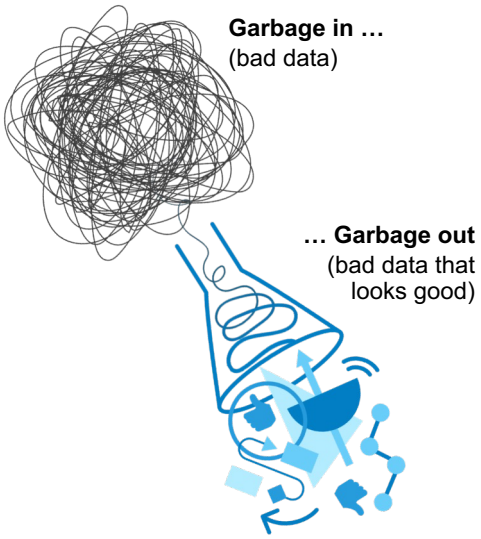
- **Scenario:** You're building a music playlist generator for a global audience
- **Questions to consider:**
  - If your training data only includes English-language songs, what experience will users in Mexico or Korea get?
  - What happens if all the data comes from people aged 20 – 30?

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Data: The Foundation That Matters

The essential truth about AI

- Despite all the headlines about AI breakthroughs, its real value still depends on one thing: data.
- The garbage in, garbage out principle:
  - Good data = smart, fair, and valuable AI
  - Bad data = biased, misleading, or even harmful AI



15

Bias, Hallucinations, and the Human Imperative

Data Bias

This is when the AI produces skewed results that reinforce stereotypes learned from its training data.

- **For example:** Asking an image AI for a "photo of a CEO" or a "nurse" often reveals the common stereotypes present in the data it was trained on.

Hallucinations

This is when the AI generates plausible but entirely false information, like inventing facts, sources, or data.

- **A real-world example:** A lawyer used an AI for legal research, and it invented several non-existent legal cases. The lawyer was later sanctioned by the court for citing them.

Human Imperative

Your expertise is the essential filter. Your professional mandate is to:

- **Validate:** Independently verify facts and claims.
- **Challenge:** Actively look for and correct bias in the output.
- **Own the outcome:** You are accountable for the final work.

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Good Data vs. Bad Data in Action

A tale of two models

- **Success story:** An ML model trained on diverse, labeled images correctly classifies wildlife photos
- **Cautionary tale:** A DL model trained only on English-language resumes shows bias against non-Western names
- **Discussion starter:** What unintended consequences could come from having bad data in your work?

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Bad Data, Different Lenses

10 min 

Group activity

- **Step 1:** Assign roles at your table (4–5 people):
  - **Data Scientist** (cares about methods & accuracy)
  - **End User** (cares about fairness & usability)
  - **Regulator** (cares about safety & compliance)
  - **Business Leader** (cares about costs & reputation)
  - *(Optional extra role if group is big: AI Ethicist)*
- **Step 2:** Choose a real-world AI use case (e.g., Netflix recommendations, facial recognition, loan approvals).
- **Step 3:** Ask the LLM: *"What could go wrong if the data for this use case is biased, incomplete, or outdated?"*
- **Step 4:** Each role responds from their perspective:
  - What's the biggest risk or harm?
  - What would you do about it?

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Debrief

3 min

Class discussion

- What was the most significant point of agreement or disagreement between the different roles at your table?
- What assumptions were challenged or uncovered when you had to discuss from your specific role's point of view?
- Based on this exercise, where are organizations most often blind when it comes to data risks, and why do those blind spots persist?

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Insights and Action

1 min

Reflect

Think about a recent AI tool you've used (ChatGPT, Netflix recommendations, etc.). What assumptions might that AI be making about you based on limited or biased data? Write down one example where 'garbage in, garbage out' could impact a decision that matters to you personally or professionally.

Apply

Ask AI: "I'm evaluating an AI hiring tool that screens resumes. The training data comes from our company's past 5 years of successful hires who were 80% male and mostly from 3 universities. Walk me through 3 specific ways this could create biased outcomes and suggest 2 concrete steps to improve the data quality before deployment."

Explore

MIT Research on AI Bias Reduction—latest bias mitigation techniques

NIST Report on AI Bias Sources—comprehensive bias framework

Harvard Business Review on Data for AI—enterprise data quality priorities

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2.3 Large Language Models

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Language Challenges for Machines

3 min

Class discussion

How do you think a computer knows that "bank" means something different in "riverbank" vs "bank account"?

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What Are LLMs and Why They Matter

- **Large Language Models (LLMs):** advanced AI systems that ***process*** and ***generate*** human language
- **They're powered by Natural Language Processing (NLP):** the field that helps machines work with text, speech, and meaning
- **The big shift:** From programming language rules manually to training models that *learn language* from examples

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So, how do Large Language Models work with human language?

We must translate our text into something the machine understands…. Numbers.

This process is the starting point for everything an LLM does.

2-25

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How Language Is Translated into Numbers

Tokenization

1 Words in a Sentence — Raw Text

Once upon a time

2 Broken It Into Pieces — Tokens

`Once` `upon` `a` `time`

Encoding

3 Each Piece Gets a Number — Token IDs

`18049` `7557` `261` `1058`

4 Each Number Gets a Location — Embeddings

Word	Token ID	Relationship Coordinates		
Once	18049	0.98	-0.19	….
upon	7557	-0.91	0.34	….
a	261	0.51	0.41	….
Time	1058	0.23	-0.67	….

(examples)

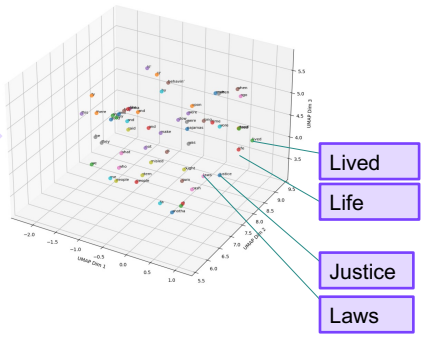
26

How These Numbers Create Meaning For The LLM

Each Number Gets a Location — Embeddings

Word	Token ID	Relationship Coordinates		
Once	18049	0.98	-0.19	….
upon	7557	-0.91	0.34	….
a	261	0.51	0.41	….
Time	1058	0.23	-0.67	….
…..	…..	…..	…..	…..

Visualizing These Locations — Vector Space



Key Insight: These aren't random numbers—they're *positioned* in a space where similar relationships are near each other.

AI-Native Training / © Scaled Agile, Inc.

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LLMs Use These Numbers to Predict the Next Token

Input: Prompt

Token	Relationship Coordinates		
Once	0.98	-0.19	….
Upon	-0.91	0.34	….
A	0.51	0.41	….
___	___	___	….

→



THE MODEL

Finds patterns, predicts next Token

→

Output: Model Prediction

Token	Relationship Coordinates		
Once	0.98	-0.19	….
Upon	-0.91	0.34	….
A	0.51	0.41	….
Time	___	___	….

The model's entire job: Look at the input, find patterns from training, and predict the most statistically likely next Token. Then repeat—one Token at a time—until the response is complete.

AI-Native Training / © Scaled Agile, Inc.

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How LLMs Are Created: Two Phases

Phase 1: Pre-Training

- Trained on massive datasets
- Learns general language patterns
- Predicts next token in context
- Mostly self-supervised

THE RESULT: AI Learns Language.
Only knows how to "predict the next Token"

User: What is 1 + 1?
AI: +1?

Phase 2: Fine-Tuning

- Adapts to follow instructions
- Human Feedback (RLHF)
- Can include domain-specific tuning
- Refines tone, truthfulness, alignment

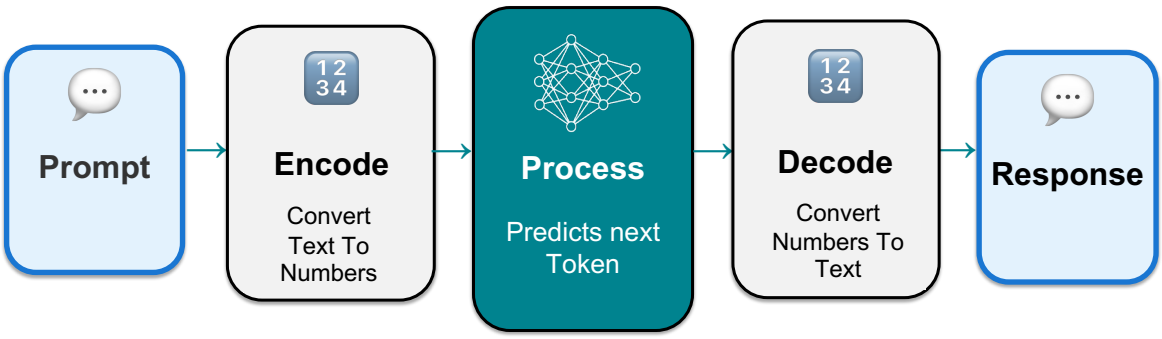
THE RESULT: AI Learns Behavior.
Trained to be a helpful assistant

User: What is 1 + 1?
AI: Great question! 1+1 equals 2. Is there anything else I can help with?

AI-Native Training / © Scaled Agile, Inc.

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Querying an LLM: The Full Loop



AI-Native Training / © Scaled Agile, Inc.

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Why This Matters

WHAT YOU NOW KNOW

- Words become numbers with locations
- The model predicts one token at a time
- It learned language, then learned to be helpful
- It's not magic—it's a probability engine

WHAT THIS MEANS FOR YOU

Hallucinations
"Likely" ≠ "True"— always verify

Prompting
Better prompts = better predictions

Context
More context = smarter responses

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Think Like a Language Model – Round 1

4 min

Group activity

- **Round 1: Warm-up Guess**
  - As a group, complete this sentence:
    - "The cat sat on the ___".
  - Each group writes down 3 guesses for the next word.
  - Reveal the LLM's top 5 probable next words with % likelihoods.

Round 1: The Cat Sat On The ...

	Next Word	Probability
1	mat	62.0%
2	floor	18.0%
3	sofa	8.0%
4	ground	7.0%
5	couch	5.0%

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Think Like a Language Model – Round 2

4 min

Group activity

- **Round 2: Longer Context**
  - As a group, complete this sentence:
    - "On a rainy evening, the cat curled up next to the warm fireplace. The cat sat on the _____".
  - Repeat the guessing game.
  - Compare how context changed the probabilities.

Round 2: On A Rainy Evening, The Cat Curled Up Next To The War...

	Next Word	Probability
1	rug	45.0%
2	floor	21.0%
3	mat	15.0%
4	cushion	10.0%
5	carpet	9.0%

Think Like a Language Model – Round 3

4 min

Group activity

- **Round 3: Open-Ended Creative Prompt**
  - As a group, complete this sentence:
    - "The CEO stormed into the room. The meeting was _____"
  - Groups make predictions, then see how probabilities spread across serious vs. humorous vs. neutral continuations.

Round 3: The CEO Stormed Into The Room. The Meeting Was ...

	Next Word	Probability
1	tense	34.0%
2	canceled	22.0%
3	chaotic	18.0%
4	productive	14.0%
5	over	12.0%

Debrief

3 min

Class discussion

- How did adding more context in each round change your predictions?
- What risks does this game reveal about an AI misinterpreting user intent?

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Insights and Action

1 min

Reflect

Think back to the “Be the Model” activity. What did you notice about how you predicted the next word? In two minutes, write about what this taught you about how LLMs work—and how your own assumptions shaped the prediction process.

Apply

Ask AI: “I’m going to give you a sentence with the last word missing: ‘The weather forecast says it will be sunny, so I should pack my ___.’ Before you complete it, walk me through your thinking process. What context clues did you use? What other words could fit? How does the earlier part of the sentence influence your choice?”

Explore

TikToken Visualizer—interactive tokenizer demo
IBM on Context Windows—why context matters in AI
Stanford CS Course Materials—LLM architecture basics

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2.4 Prompting and Safe Use

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Prompt Insights

5 min

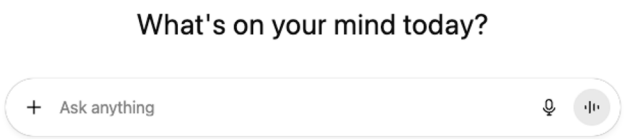
Pair share

- **Scenario:** Have you ever typed something into ChatGPT or another AI tool and been disappointed or impressed by what you got back?
- **Instructions:**
  - In pairs: One person shares a prompt that worked really well or one that totally missed the mark
  - Discuss what might have made the difference
  - Switch up and repeat

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What Exactly Is a Prompt?

- A prompt is the set of instructions you feed an AI system—your leverage point for steering its response
- Effective prompts specify objective, scope, and tone, just as a brief would for a trusted analyst
- Think of it as assigning a task to a high-performing intern—clarity in, clarity out



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Prompting Techniques – Start Here

GOAL	Define the end objective of the prompt	*Draft a marketing plan for a new product launch*
ROLE	Instruct the AI to act as a specific expert or persona	*Act as a seasoned financial analyst reviewing a quarterly report*
TASK	Describe the action you want the AI to take	*Summarize the attached report into three key takeaways for an executive audience*
CONTEXT	Provide AI with specific background knowledge not in its training data	*Our company is a startup in the renewable energy sector targeting small businesses*
DETAILS	List instructions on formatting, limitations, guardrails, style, success criteria, etc.	*Write in a professional but friendly tone, and limit the response to 300 words*

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Intermediate Prompting Techniques

EXAMPLES	Provide examples to help it generate responses in the right style, tone or format.	*Here is an example of a good customer service email: [insert example]. Write a response to an unhappy customer in the same style*
DATA	Feed AI structured data (e.g., text, CSV, JSON) to analyze, synthesize, etc.	*Analyze the uploaded CSV file of sales data and identify the top-performing region*
INTERACTION	Instead of getting one-shot response, instruct AI to ask clarifying questions.	*I need a compelling product description. Ask me clarifying questions about the product before you create it.*
STEPS	Decompose complex prompts into smaller logical steps	*First, summarize the Q3 financial report. Second, compare it to Q2. Finally, list three areas for improvement.*
REASONING	Ask AI to explain why it gave a response, or critique itself and improve its answer based on its assessment.	*List the top 3 new product opportunities for our company. Then, critique your own list and suggest an even better alternative based on that critique.*

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Advanced Prompting Techniques

INTERVIEW	Instruct AI to conduct a dialogue with you before completing additional tasks	*I want to improve employee morale. Interview me about the current company culture and challenges before suggesting solutions.*
CHALLENGE	Provide a solution, an idea, etc. and asking AI to critique it and identify improvements	*Here is our current marketing slogan: [insert slogan]. Point out potential weaknesses and suggest three stronger alternatives.*
DECOMPOSITION	Instead of you specifying steps, task AI to define the necessary steps to achieve an objective	*I want to expand our business into the European market. What are the necessary steps to plan for a successful launch?*
ANALYSIS	Provide data that AI explores freely without specific instruction other than finding patterns, insights, trends, etc.	*Analyze this customer feedback data and identify any surprising patterns or emerging trends in customer sentiment.*
CHAIN-OF-THOUGHT	Instead of giving just an answer, ask AI to show its reasoning process (steps, input, extensive output) – deep reasoning capability	*Determine the best city for our new office based on these criteria: [list criteria]. Show your step-by-step evaluation for each city.*

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Why Precision Prompting Pays Off

- **Quality**: Detailed prompts yield responses that need less post-editing
- **Velocity**: Clear direction accelerates ideation, analysis, and decision cycles
- **Risk management**: Well-structured prompts reduce the odds of hallucinations or off-brand language
- **Accountability**: The human prompter remains the final arbiter—review, revise, and own the outcome



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Understanding the Risks of AI in the Workplace

Data Privacy & Confidentiality

- Submitting confidential company data, client information, or Personal Identifiable Information (PII) to AI can lead to a data breach. It can also violate data privacy laws.

Accuracy & "Hallucinations"

- AI models can generate incorrect, misleading, or completely fabricated information, but present it with complete confidence as factual.

Inherent Bias

- AI is trained on vast amounts of data from the internet, which contains human biases. It can unintentionally reproduce or even amplify these biases related to race, gender, age, and more.

Intellectual Property (IP) & Copyright

- Using AI to generate content could infringe on existing copyrights if the model was trained on protected material. Ownership of AI-generated content can also be ambiguous.



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Responsible AI: Global Standards & Governance

The Emerging Regulatory Landscape: Major governing bodies are creating frameworks to guide AI development and use. Your company's AI policy will likely be based on these emerging standards.

- 🇪🇺 **The EU AI Act:** A landmark law that classifies AI systems based on risk (from minimal to unacceptable) and imposes rules accordingly.

  https://artificialintelligenceact.eu/

- 🇺🇸 **NIST AI Risk Management Framework:** A voluntary framework from the U.S. National Institute of Standards and Technology that provides a structured process for managing AI-related risks.

  https://www.nist.gov/itl/ai-risk-management-framework

- 🌐 **ISO/IEC Standards (e.g., ISO/IEC 42001):** International standards that help organizations responsibly manage the entire lifecycle of an AI system.

  https://www.iso.org/standard/42001



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Your Guide to Safe & Ethical AI Use

✅ **Protect Information: Assume It's Public.**

NEVER input confidential company data, trade secrets, financial records, employee information, or client PII into a public AI tool. Use only company-approved, private AI instances for sensitive work.

✅ **Verify, Don't Trust: Be the Skeptic.**

Treat AI-generated content as a **first draft**, not a final product. Always fact-check data, statistics, and critical claims before using them. Demand evidence from the AI. You are responsible for the final output.

✅ **Follow Company Policy: Know the Rules.**

Understand and adhere to your company's specific guidelines on using AI tools. Which tools are approved? How should you disclose AI-generated content? When in doubt, ask.

✅ **Review for Bias or Harmful Content: Think Critically.**

Carefully review text and images for subtle biases, stereotypes, or any other language that could harm people, organizations, or society. If the output feels "off" or unfair, challenge it and refine your prompt or approach.

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Your Guide to Safe & Ethical AI Use (cont'd)

✅ **Be Transparent: Disclose When Necessary.**

Be honest with your team and your clients about the role AI played in your work, especially for substantive contributions. Don't pass off AI-generated work as entirely your own.

✅ **Ask for the Process.**

Instruct AI to show its work so you can validate it. Use prompts like: "Explain your reasoning step-by-step."

✅ **Remember AI is Everywhere!**

These rules don't just apply to chatbots. Be mindful when using AI-powered features in other software, such as:

- Email assistants (e.g., summarizing threads, drafting replies)
- Meeting software (e.g., AI-generated summaries and action items)
- Code completion tools and document editors with "smart" features.
- AI features built into traditional software systems (office productivity, large corporate systems)

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Know Your Models*

Model Name	Developer	Version Date	Best Suited For
Gemini 3	Google DeepMind	**Nov 18, 2025**	Multimodal + stronger reasoning/coding; available across Google products (per Google's Gemini 3 announcement).
GPT-5.2	OpenAI	**Dec 11, 2025**	Professional knowledge work + long-running agents (per OpenAI's release post).
Claude Sonnet 4.5	Anthropic	**Sep 29, 2025**	Practical frontier performance with emphasis on coding and general AI assistant workloads (per Anthropic's Sonnet model page).
Grok 4.1	xAI	**Nov 17, 2025**	General assistant use; updated flagship model available in Grok apps and model picker (per xAI announcement).
Llama 4	Meta	**Apr 5, 2025**	Open (weights available), **natively multimodal** model family (per Meta's Llama 4 announcement).
DeepSeek-V3.1	DeepSeek	**Aug 28, 2025**	Large-context model offering (e.g., Vertex AI lists **163,840** context length and release date).
Mistral Small 3.2	Mistral AI	**Jun 20, 2025**	Small model update with **128k** context (per Mistral model page).
Falcon 3	Technology Innovation Institute (UAE)	**Dec 17, 2024**	Open-source Falcon 3 family (per TII release announcement).
Qwen3-Max	Alibaba Cloud	**Sep 24, 2025**	Large LLM positioned for code generation and agentic capabilities (per Alibaba Cloud roadmap post).

* Models are updated frequently, and capabilities change – this list will be updated periodically

See How Prompts Change Responses

7 min

Group Activity

- **Step 1:** As a group, pick a work-relevant question (e.g., *"How can we use AI to improve customer support?"*).
- **Step 2:** Ask the LLM to help using **techniques from this lesson**:
  - **Round 1 –** Just use basic prompt techniques
  - **Round 2 –** Try 1 or 2 intermediate techniques
  - **Round 3 –** Try 1 or 2 advanced techniques
- **Step 3:** Compare and discuss the responses from each round.

Debrief

3 min

Class Discussion

- Which of the prompting techniques gave you the most useful response, and why?
- When would you choose to use a specific technique?
- Based on this, what's your number one tip for someone frustrated with the quality of their AI results?

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Insights and Action

1 min

Reflect

Think about a time when an AI gave you a response that felt off. In two minutes, write what you think went wrong—was it the prompt, the model, or your expectations? How will you approach prompting differently now?

Apply

Ask AI: “Using the RISE framework, write a clear and effective prompt to help me summarize key decisions from a recent team meeting. Include Role, Input, Steps, and Expectation.”

Explore

Google DeepMind Prompt Engineering Guide—Detailed Prompting Techniques

AI LLM Leaderboard—Compare any two models against each other

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2.5 Retrieval Augmented Generation (RAG)

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Can ChatGPT Answer This?

3 min

Class discussion

How well do you think ChatGPT can answer these without help?

- “What’s our internal PTO policy?”
- “What is the difference between LLM and RAG?”
- “Who leads cybersecurity here?”

The Problem RAG Solves

When AI confidence becomes risk

- **The scenario:**
  - ChatGPT says: “You get unlimited vacation”
  - HR policy says: “10 days”
- **Question:** Who’s responsible if that misleads an employee?
- **The core issue:** “When AI gives confident answers—but they're wrong—it creates risk, not value.”



What Is RAG?

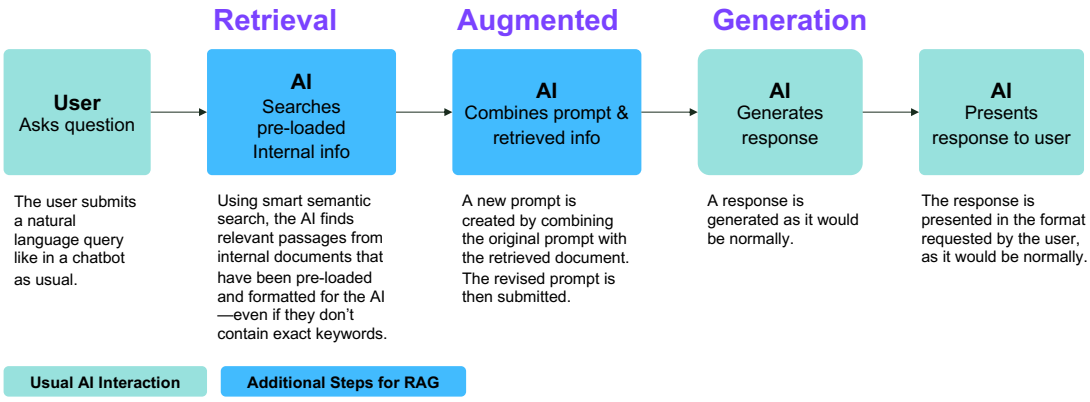
Retrieval Augmented Generation: 'AI with receipts'

- **Definition:** The model doesn't just make things up; it looks things up."
- **The difference:**
  - LLMs alone use static training data, and with fine-tuning, infrequently updated data
  - RAG retrieves current, specific information from trusted sources at the time of the question
  - If you've ever uploaded a PDF file in a ChatGPT window before asking it a question, that's RAG

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How Does RAG Work?

- **Scenario:** You ask your AI assistant, "What's our policy for international remote work?"



- **Why it matters:** RAG keeps responses grounded in real company knowledge—reducing hallucination risks and improving trust, accuracy, and compliance.

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When to Use RAG

RAG shines when

- Information changes frequently
- Precision matters
- You need source-grounded answers

Real-world use cases



Internal help desks



Customer service chatbots



Employee onboarding assistants



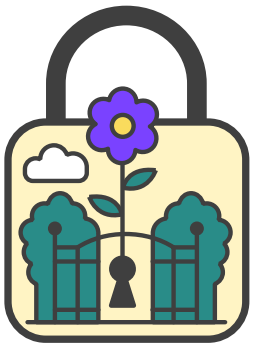
Field support tools

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The Value of a Gated Garden

RAG is only as good as its data sources

- What's a Gated Garden? Your organization's secure, curated source of truth (employee handbook, SOPs, knowledge base, contracts)
- Benefits to employees:
  - Get answers faster, with less doubt
  - Rely on one consistent source, not Slack or old PDFs
- Benefits to organizations:
  - Fewer repeated questions
  - Reduced risk of misinformation
  - Stronger trust in internal tools



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RAG vs. Fine-Tuning

Two ways to make AI smarter

1. Provide specific knowledge in documents: **RAG**
   - Keeps knowledge separate from model
   - Can update frequently as sources change
   - Maintains clear source attribution
2. Retrain the model itself: **Fine-Tuning**
   - Bakes specific knowledge into the model itself
   - Requires training the model with domain knowledge (time and cost)
   - Loses source attribution

> **Key insight:**
> RAG is usually better for factual knowledge; fine-tuning for style and behavior

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Basic Prompt vs. RAG: Spot the Difference

7 min

Group activity

- **Step 1 — Assign Personas**
  - **HR Manager** → Employee Handbook
  - **Finance Analyst** → Expense CSV (budgets vs. actuals)
  - **Sales/CS Manager** → Product FAQ / Playbook
  - **Product Manager** → Release Notes / Technical Docs

Download reference files for each persona in your Companion App

- **Step 2 — Ask the Same Question Twice**
  - **Basic LLM Prompt:** Ask a question related to your persona without documents.
  - **RAG Prompt:** Ask the exact same question, but instruct:

    *"Answer **only** using the provided document [handbook/CSV/FAQ/docs]. Do not guess or add outside info." (upload your document with this prompt)*
- **Step 3 — Compare & Capture**
  - Where was the basic answer vague, generic, or wrong?
  - Where did the RAG answer add precision, facts, or structure?
  - Record **1 "RAG win"** (where grounding helped) and **1 remaining limitation**.

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Debrief

3 min

Class discussion

- Where did grounding the model in your document make the biggest difference?
- How did this affect your confidence in making a business decision based on the AI's answer?
- Where else in your organization could this RAG approach be a game-changer?

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Insights and Action

1 min

Reflect

Think about the difference between a model that 'knows' vs. one that "'ooks it up.' In two minutes, write about a work scenario where relying on outdated or general AI knowledge could create risk—and how RAG could help solve that.

Apply

Ask AI: “I’m designing a chatbot for employee HR questions. What are 3 ways I can use Retrieval Augmented Generation (RAG) to make sure the answers are accurate, up-to-date, and traceable?”

Explore

How RAG Works—Explained by IBM

RAG vs Fine-Tuning vs Prompt Engineering—13 min video on how to get better output from LLMs

Real World RAG Examples—10 actual use cases of RAG in real companies

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2.6 | AI Agents

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Class Schedule

Day 1

Lesson 1: EDGE & AI-Native

- 1.1 | The EDGE World
- 1.2 | AI-Native

Lesson 2: AI Technical Literacy

- 2.1 | AI Basics
- 2.2 | Data
- 2.3 | Large Language Models
- 2.4 | Prompting and Safe Use
- 2.5 | Retrieval Augmented Generation (RAG)
- 2.6 | AI Agents
- 2.7 | Frontiers In AI

Day 2

Lesson 3: The AI-Native Success Factors

Lesson 4: Improve My Workflow With AI

Lesson 5: AI as Your Thought Partner

Lesson 6: Roadmap to AI-Native

Lesson 7: Tell Your AI-Native Story

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If You Could Clone Yourself

3 min

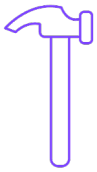
Class discussion

> Can you think of a time in a film or TV series when a helpful AI or robot really supported someone in a meaningful way?

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What Are AI Agents?

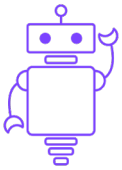
The evolution of Technology assistance



- **Tools:** Single-purpose; you drive (calculator, spell-check)



- **Assistants:** Multi-capable; you guide (Siri, Alexa)



- **Agents:** Goal-oriented; you direct, autonomous systems that take initiative (Self Driving Car)

> **The key difference:**
> Agents can plan, execute, and adapt without constant supervision

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Core Agent Capabilities

Four core capabilities:

1. **Goal understanding:** Can interpret what you want accomplished
2. **Planning:** Can break down complex tasks into steps
3. **Tool use:** Can leverage other systems to get things done
4. **Adaptation:** Can handle unexpected situations and course-correct

> **Think of it like this:**
> The difference between a GPS that just shows a map (tool) versus a self-driving car that gets you there (agent)

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Different Agents For Different Needs

- **Research agents:** Find, synthesize, and summarize information
- **Creative agents:** Generate and refine content based on guidelines
- **Workflow agents:** Coordinate tasks across systems and people
- **Decision support agents:** Analyze options and recommend action

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Agent Differentiation: Real-world Examples

- **Research agent:** An AI-powered literature review assistant that scans hundreds of articles and summarizes the five most relevant findings for a student writing a paper
- **Creative agent:** An AI content helper that rewrites a company's blog post to match brand voice and tone, following SEO guidelines and editorial rules
- **Workflow agent:** A Slack-integrated assistant that monitors team flow and sends reminders to teammates when deadlines approach or blockers arise
- **Decision support agent:** An AI tool that compares three insurance plans based on your budget and coverage needs, explains the pros and cons, and recommends the best fit

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Strategic Value Across Company Types

Finding the right balance

- **Human strengths:**
  - Judgment and ethics
  - Creative direction
  - Contextual understanding
  - Final decision-making
- **Agent strengths:**
  - Information processing at scale
  - Consistent execution
  - 24/7 availability
  - Handling routine complexity

Key principle:
Agents should amplify human capability, not replace human judgment

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Mental Model For Understanding AI Agent Terminology

Agents Are Evolving From A Single Assistant Into An Organized Workforce.

AI Agent

A single, capable agent that can understand and execute commands.

Multi-Agent Orchestration

The coordination of a team of agents, managing how they work together to achieve a common goal.

Agentic AI

A proactive agent who takes initiative and can take autonomous actions.

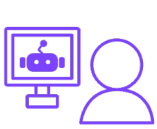
Agentic Workflow

A structured process where multiple agents collaborate to deliver a specific outcome.

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Multi-agent Orchestration: LinkedIn Article Writer

The power of agents when they work together to achieve a greater goal



Post a weekly AI thought leadership article

Orchestration

Oversees the full process from idea to publication
- Triggers each agent in sequence
- Checks output quality at each step
- Handles exceptions or retries
- Passes instructions and previous agent's output to next agent

Agent	Trend scout	Ideation	Validation	Outline	Writing	Style & tone	Schedule	Publish
Role	Identify 3 – 5 trending topics from LinkedIn, news, and AI blogs	Brainstorm 3 – 5 potential article ideas (Title and one-line summary) based on chosen trend	Rank ideas by novelty, clarity, and relevance to the target audience (e.g., execs, AI strategists)	Create a detailed outline: Headings Intro Body sections Call to action	Write a 600- to 800-word first draft of the article based on the outline	Refine voice for LinkedIn (e.g., professional, conversational, thought-leadership tone)	Determine optimal posting time based on audience engagement patterns	Post the article to LinkedIn via API (or cues for human approval)

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Agentic Workflow: Meal Recommender (Part 1)

15 min

Individual activity

- **Step 1:** Ensure you have set up accounts for **Flowise** and **Groq** (get your Groq API key).
- **Step 2:** In Flowise, go to Agent Flows → Add New (a Start node appears).
  - Add an Agent node and set Provider: Groq → paste API key → select a model.
  - Configure and run the Agent (Instructions and prompt in the companion app)
  - Run the Agent inside Flowise to test it

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Agentic Workflow: Meal Recommender (Part 2)

15 min

Individual activity

- **Step 3:** Add a tool
  - Add the **Calculator** tool (built-in) and **connect** it so the Agent can call it.
  - Update the **agent prompt** to require calculations
  - Run the Agent inside Flowise to test it
- **Step 4:** Share your AI agent with the world
  - Click on “Share Chatbot” in the API Endpoint Menu
  - Make the Endpoint Public
  - Copy the public URL to share your new agent

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Debrief

3 min

Class discussion

- Reflect on the ability to create agents without programming knowledge. What does this unlock for you?

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Insights and Action

1 min

Reflect

Think about a recent work situation where you felt overwhelmed by repetitive tasks or complex coordination. What would change about how you approach your work if you shifted from being someone who uses AI tools to someone who directs AI agents? Write about this transition—what excites you most and what concerns would you want to address about maintaining the right balance between human judgment and agent autonomy?

Apply

Ask AI: "I want to design an AI agent for my workplace. Help me think through this step-by-step: First, identify a recurring workflow in my job that involves multiple steps and decisions. Then, break down which parts an AI agent could handle autonomously (goal understanding, planning, tool use, adaptation) versus where I need to maintain human oversight. Finally, describe what inputs the agent would need, what it would produce, and what boundaries I should set. Make this specific to a real work challenge I face."

Explore

IBM – 2025 The Year of AI Agent Exploration—expectations vs. reality

Open Source Agentic Frameworks—as of April 2025

Boost productivity with AI agents (IBM)—access the Gartner report

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2.7 Frontiers in AI

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Lesson 3: The AI-Native Success Factors

Lesson 4: Improve My Workflow With AI

Lesson 5: AI as Your Thought Partner

Lesson 6: Roadmap to AI-Native

Lesson 7: Tell Your AI-Native Story

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Where Are We Headed?

3 min

Class discussion

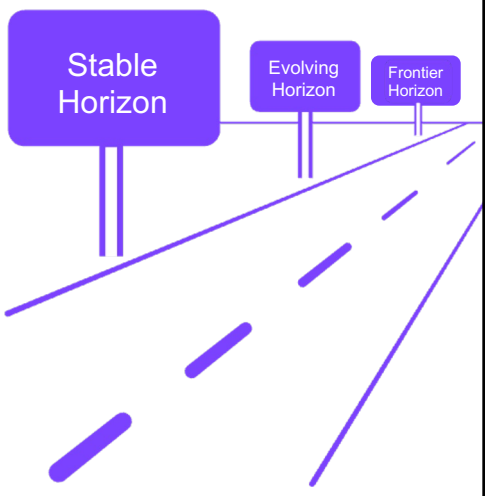
- Think about your phone: What's something it can do now that felt experimental or futuristic just a few years ago?
- What do you imagine it might be able to do five years from now?

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AI Horizons by Capability

1. **Stable horizon:** Reliable, production-ready capabilities
   - Text generation, image creation, classification, summarization
2. **Evolving horizon:** Working but still improving capabilities
   - Multi-step reasoning, code generation, specialized knowledge work
3. **Frontier horizon:** Emerging or experimental capabilities
   - General reasoning, multimodal creation, autonomous agents

Key principle: Match your expectations and implementation plans to the right horizon



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Extra, Extra, Read All About It!

12 min

Group activity

- **Step 1:** Pick any future year—realistic or bold.
- **Step 2:** Write a headline that starts with: The year is [_____] ... and AI has become our everyday partner in _____.
- **Step 3:** Bring your story to life visually
  - Under your headline, draw or sketch a simple visual story that shows the world this AI has helped shape. Include:
  - How has it changed people's lives or work?
  - Any surprising outcomes, innovations, or new needs that emerged?
- **Step 4:** Share
  - Each group presents their headline and 'breaking news' to the room.

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Debrief

3 min

Class discussion

- What themes did you notice across the different future headlines?
- Which table's scenario felt most realistic vs. most ambitious?
- What would need to happen between now and these futures to make them real?

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Insights and Action

1 min

Reflect

Consider an AI capability you're currently using or considering for your work. Based on what you learned about the three horizons, honestly assess where this capability sits today: Is it Stable (ready for full reliance), Evolving (worth experimenting with), or Frontier (experimental only)? Write down how this classification should change your approach and expectations when using it.

Apply

Ask AI: "Help me categorize these AI capabilities for [my] industry: automated customer service responses, AI-powered data analysis, predictive maintenance, autonomous decision-making systems, and AI-generated creative content. For each one, tell me if it's Stable, Evolving, or Frontier, and give me one specific recommendation for how I should approach implementing it based on its maturity level."

Explore

MIT enterprise AI maturity research

Cisco global AI readiness assessment

NIST AI risk management standards

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Lesson Review

You can:

- Navigate common AI terminology
- Understand how Large Language Models work
- Craft effective prompts
- Apply the principles of Responsible AI
- Describe Retrieval-Augmented Generation (RAG)
- Understand how AI Agents work and their capabilities
- Classify AI use cases as Stable, Evolving, or Frontier

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